An information theoretic view on selecting linguistic probes

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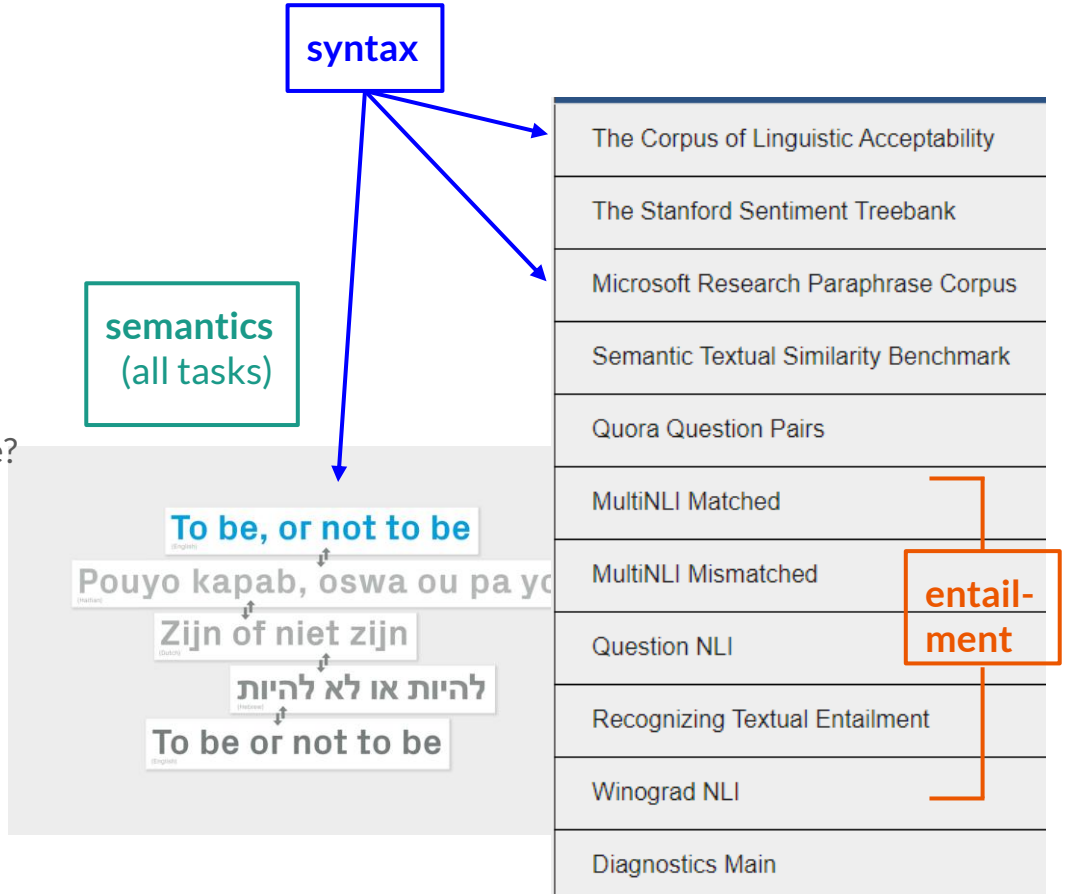
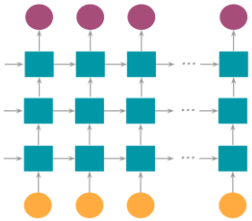
Introduction

- Neural NLP models takes over SOTA
- Why can neural NLP models do so well?
- They have linguistic knowledge.
- If yes, how much knowledge do they have?
  - Where do they encode the knowledge?
  - Do they encode only in some neurons?
  - Patterns of encoding knowledge?



Diagnostic classifiers

- (Ettinger et al., 2016): Diagnostic classifier task
  - Probe the sentence representations.
- (Alain & Bengio, 2017):
  - Diagnostic classifier essentially probes: *"Is there information about factor ___ in this part of the model?"*



Probing for semantic evidence of composition by means of simple classification tasks

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Understanding intermediate layers using linear classifier probes

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What knowledge to probe?

- Many types of probing tasks:
  - Syntax distance (Hewitt and Manning, 2019)
  - Syntax & semantic tasks (Tenney et al., 2019)
  - Rhetorical discourse features (Zhu et al., 2020)
  - Many other knowledge.
- Diagnostic classifier:
  - Simple set-up.
  - Good performance -> rich knowledge.

A Structural Probe for Finding Syntax in Word Representations

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BERT Rediscovers the Classical NLP Pipeline

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Examining the rhetorical capacities of neural language models

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A dichotomy about probe

- (Hewitt and Liang, EMNLP 2019)
- When we observe good performance:
  - Do the representations contain rich knowledge?
  - Or, do the probe learns the task?
  - (Zhang et al., ICLR 2017): NNs can learn even from random vectors!
- Propose: select probes using “selectivity” criterion.
  - Selectivity: how much the probing accuracy *improves* compared to the control task (random labels).
- Propose to use the probes with as few parameters as possible.

UNDERSTANDING DEEP LEARNING REQUIRES RETHINKING GENERALIZATION

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Designing and Interpreting Probes with Control Tasks

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An info-theoretic formulation

- The purpose of a diagnostic classifier is to approximate I(T;R)
  - T: the label
  - R: the representation
- Reject the "good representation or good probe" dichotomy
- Reject the *control tasks*
  - Propose to use *control function* (randomize R) as a control setting.
  - With control function, compute the "information gain" criterion to select probes.
  - Info gain: the difference of cross-entropy losses between control task and probing task:

$$\tilde{\mathcal{G}}(T, R, \mathbf{c}) = H(p_c, q_{\phi_c}) - H(p, q_\phi)$$

Information-Theoretic Probing for Linguistic Structure

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An info-theoretic view on the dichotomy

- The dichotomy is valid, info-theoretically.
- Decompose the good probing performance:
  - T: target. R: representation.
  - p: unknown true distribution; q: the probe model

$$H(p, q_\theta) = H(T) - I(T; R) + \mathrm{KL}(p \,\|\, q_\theta)$$

A low cross-entropy "probing loss" could be the results of:

(a) High code-target mutual information ("good representation")

Or (b) A low KL ("probe learns the task")

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An info-theoretic view on selecting probes

- Selecting probes with the two controlling mechanisms:
  - *Control task* (Hewitt and Liang, 2019)
  - *Control function* (Pimentel et al., 2020)
- They still contain errors, but much smaller:
  - The error terms are both the diff of a pair of KL divergences
- They differ by only irrelevant terms!
  - … as long as the randomization is done well
  - How about empirically?

$$H(p_c, q_{\theta_c}) - H(p, q_\theta) = I(T; R) - \Delta$$

$$\Delta_h = \mathrm{KL}(p \,\|\, q_\theta) - \mathrm{KL}(p_c \,\|\, q_{\theta_c}) + \mathrm{Const}$$

$$\Delta_p = \mathrm{KL}(p \,\|\, q_\phi) - \mathrm{KL}(p_c \,\|\, q_{\phi_c})$$

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The two control mechanisms agree well

- Ran 10,000+ POS probing experiments, sweeping different param configs.
- Empirically: the criteria of Hewitt and Liang (2019) and Pimentel et al., (2020) agree,
  - To the extent similar to the "accuracy vs. cross entropy loss" agreement

Language	# POS	# Tokens train / dev / test	Correlations (t_acc,f_ent)	(t_acc,t_ent)	(f_acc,f_ent)
English	17	177k / 22k / 22k	0.1615	0.1334	0.1763
French	15	303k / 31k / 8k	0.0906	0.0606	0.1295
Spanish	16	341k / 33k / 11k	0.1360	0.0560	0.1254

Table 1: Spearman correlations between t_acc (the "selectivity" criterion (Hewitt and Liang, 2019)) and f_ent (the "gain" criterion (Pimentel et al., 2020)) are on par with two "accuracy vs. cross entropy" correlations.

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Takeaways

- "Diagnostic classifier" probes can be formulated better with information theory.
  - We analyzed the sources of error of (1) single loss, (2) control mechanisms.
  - We showed the two control mechanisms are equivalent.

Special thanks to:

Mohamed, Amanjit, Bai, Yuchen, Chuer, Aparna, Frank -- for discussions about the ideas + relevant papers